

**University of Toronto Mississauga**  
**Department of Mathematical and Computational Sciences**

**Applied Time Series Analysis**  
**STA457H5S, Winter 2021**  
**Dr. Masoud Ataei**

## Test 1 Solutions

**Question 2 (20pts).** Let  $\{X_t\}$  be a stochastic process defined by

$$X_t = U \cos(\theta t) + V \sin(\theta t)$$

where the parameter  $\theta \in [-\pi, \pi]$ , while  $U$  and  $V$  are two uncorrelated random variables such that

$$\mathbb{E}[U] = \mathbb{E}[V] = 0 \quad \text{and} \quad \text{Var}[U] = \text{Var}[V] = 1.$$

Derive the mean, variance, autocovariance function and autocorrelation coefficient function for the stochastic process  $\{X_t\}$ . Discuss whether this process is a weakly stationary one?

**Hint.** You may use the following trigonometric identities:

$$\sin(\alpha) \sin(\beta) = \frac{1}{2} [\cos(\alpha - \beta) - \cos(\alpha + \beta)]$$

$$\cos(\alpha) \cos(\beta) = \frac{1}{2} [\cos(\alpha - \beta) + \cos(\alpha + \beta)]$$

**Solution.**

**Mean value:**

$$\mathbb{E}[X_t] = \mathbb{E}[U \cos(\theta t) + V \sin(\theta t)] = \cos(\theta t) \mathbb{E}[U] + \sin(\theta t) \mathbb{E}[V] = 0$$

**Autocovariance function:**

$$\begin{aligned} \gamma_{t,t+h} &= \mathbb{E}\{[U \cos(\theta t) + V \sin(\theta t)][U \cos(\theta(t+h)) + V \sin(\theta(t+h))]\} \\ &= \mathbb{E}\{\cos(\theta t) \cos(\theta(t+h))U^2 + \sin(\theta t) \sin(\theta(t+h))V^2 + \cos(\theta t) \sin(\theta(t+h))UV + \sin(\theta t) \cos(\theta(t+h))UV\} \\ &= \cos(\theta t) \cos(\theta(t+h))\mathbb{E}[U^2] + \sin(\theta t) \sin(\theta(t+h))\mathbb{E}[V^2] + \cos(\theta t) \sin(\theta(t+h))\mathbb{E}[UV] + \sin(\theta t) \cos(\theta(t+h))\mathbb{E}[UV] \end{aligned}$$

Recall that two random variables are uncorrelated when their correlation coefficient is zero:

$$\rho(U, V) = \frac{Cov(U, V)}{\sqrt{Var[U]Var[V]}} = 0$$

which implies that covariance of the uncorrelated random variables  $U$  and  $V$  is zero. Then, we can obtain

$$Cov(U, V) = \mathbb{E}[UV] - \mathbb{E}[U]\mathbb{E}[V] = 0 \implies \mathbb{E}[UV] = \mathbb{E}[U]\mathbb{E}[V] = 0.$$

So, the last two terms of the autocovariance function would become zero. Furthermore, we have that

$$Var[U] = \mathbb{E}[U^2] - (\mathbb{E}[U])^2 \implies \mathbb{E}[U^2] = Var[U] = 1.$$

By the same rationale, we can write

$$\mathbb{E}[V^2] = Var[V] = 1.$$

Hence, the expression for the  $\gamma_{t,t+h}$  simplifies to:

$$\gamma_{t,t+h} = \cos(\theta t) \cos(\theta(t+h)) + \sin(\theta t) \sin(\theta(t+h)) = \frac{1}{2}[\cos(\theta h) - \cos(2\theta t + \theta h)] + \frac{1}{2}[\cos(\theta h) + \cos(2\theta t + \theta h)] = \cos(\theta h)$$

**Variance:**

$$h = 0 \implies Var[X_t] = \gamma_{t,t} = \cos(0) = 1$$

**Autocorrelation coefficient function:**

$$\rho_h = \frac{\gamma_{t,t+h}}{Var[X_t]} = \cos(\theta h)$$

Since the mean value is a constant and the autocovariance function depends only on the lag parameter  $h$ , the process is weakly stationary.

**Question 3 (20pts).** Consider an autoregressive moving average (ARMA) model given by

$$X_t = \frac{1}{3}X_{t-1} + \frac{2}{9}X_{t-2} + W_t - \frac{1}{2}W_{t-1}$$

where  $\{W_t\} \sim wn(0, \sigma_w^2)$ .

- a) Determine the order of this model, i.e., identify the values of  $p$  and  $q$  such that the  $ARMA(p, q)$  would correspond to the model given above.
- b) Write down the given ARMA model in the form of an equation  $\phi(B)X_t = \theta(B)W_t$ , where  $B$  denotes a backward shift operator, and  $\phi(B)$  and  $\theta(B)$  are the so-called autoregressive and moving average characteristic polynomials, respectively.
- c) Determine whether the process is stationary.
- d) Determine whether the process is invertible.
- e) Express the model as an  $MA(\infty)$  model using the  $\phi(B)$ ,  $\theta(B)$  and the MA operator

$$\psi(B) = \sum_{i \geq 0} \psi_i B^i.$$

That is, write down the model as:

$$X_t = \psi(B)W_t,$$

where you need to compute the coefficients  $\psi_j$ s associated to the infinite-order moving average model.

- f) By analyzing the behaviour of the obtained  $\psi$  weights, deduce the possible characteristics of this particular ARMA model.

**Solution.**

- a)  $ARMA(2,1)$
- b) The autoregressive and moving average characteristic polynomials are as follows:

$$\phi(B) = 1 - \frac{1}{3}B - \frac{2}{9}B^2$$

$$\theta(B) = 1 - \frac{1}{2}B$$

So, the ARMA model can be written as:

$$(1 - \frac{1}{3}B - \frac{2}{9}B^2)X_t = (1 - \frac{1}{2}B)W_t$$

- c)

$$\phi(B) = 0 \implies 1 - \frac{1}{3}B - \frac{2}{9}B^2 = 0 \implies B = -3, \frac{3}{2}$$

Since the magnitude of both roots is strictly greater than 1, the process is stationary.

- d)

$$\theta(B) = 0 \implies 1 - \frac{1}{2}B = 0 \implies B = 2$$

Since the magnitude of the root is strictly greater than 1, the process is invertible.

**e)**

$$\begin{aligned}
 \psi(B) &= \frac{\theta(B)}{\phi(B)} = \frac{1 - \frac{1}{2}B}{1 - \frac{1}{3}B - \frac{2}{9}B^2} \\
 &= \frac{1 - \frac{1}{2}B}{(1 - \frac{2}{3}B)(1 + \frac{1}{3}B)} \\
 &= (1 - \frac{1}{2}B)(1 + \frac{2}{3}B + \frac{4}{9}B^2 + \frac{8}{27}B^3 + \dots)(1 - \frac{1}{3}B + \frac{1}{9}B^2 - \frac{1}{27}B^3 + \dots) \\
 &= 1 - \frac{1}{6}B + \frac{1}{6}B^2 + \frac{1}{54}B^3 + \dots
 \end{aligned}$$

So, we have

$$\psi_0 = 1, \psi_1 = -\frac{1}{6}, \psi_2 = \frac{1}{6}, \psi_3 = \frac{1}{54}, \dots$$

**f)** It is seen that the  $\psi$  weights die away quickly, which indicates a stationary, invertible process.

**Question 4 (10pts).** Let  $\{W_t\}$  be a Gaussian iid white noise process, i.e.,  $\{W_t\} \stackrel{iid}{\sim} N(0, 1)$ , and define the stochastic process  $\{X_t\}$  as follows:

$$X_t = \begin{cases} W_t, & \text{if } t \text{ is even,} \\ \frac{W_{t-1}^2 - 1}{\sqrt{2}}, & \text{if } t \text{ is odd.} \end{cases}$$

**a)** Show that the process  $\{X_t\}$  is white noise, but not necessarily iid. That is, show that the considered stochastic process satisfies  $\{X_t\} \sim wn(0, 1)$ , whereas the case of  $\{X_t\} \stackrel{iid}{\sim} wn(0, 1)$  is not a valid characteristic of this process.

**b)** Find the expectation  $\mathbb{E}[X_{n+1}|X_1, X_2, \dots, X_n]$  for both odd and even values of  $n$ . What conclusion you may be able to draw on the basis of your derivations?

**Solution.**

**a)** For  $t$  even we have

$$\mathbb{E}[X_t] = \mathbb{E}[W_t] = 0$$

and for  $t$  odd

$$\mathbb{E}[X_t] = \mathbb{E}\left[\frac{W_{t-1}^2 - 1}{\sqrt{2}}\right] = 0.$$

Next, we compute the autocovariance function. If  $t$  is even we have

$$\gamma_{t,t} = \mathbb{E}[W_t^2] = 1$$

and if  $t$  is odd

$$\gamma_{t,t} = \mathbb{E}\left[\left(\frac{W_{t-1}^2 - 1}{\sqrt{2}}\right)^2\right] = \frac{1}{2}\mathbb{E}[W_{t-1}^4 - 2W_{t-1}^2 + 1] = \frac{1}{2}(3 - 2 + 1) = 1$$

Furthermore, if  $t$  is even

$$\gamma_{t+1,t} = \mathbb{E}\left[\frac{W_t^2 - 1}{\sqrt{2}}W_t\right] = \frac{1}{\sqrt{2}}\mathbb{E}[W_t^3 - W_t] = 0$$

and for  $t$  odd

$$\gamma_{t+1,t} = \mathbb{E}\left[W_{t+1}\frac{W_{t-1}^2 - 1}{\sqrt{2}}\right] = \mathbb{E}[W_{t+1}]\mathbb{E}\left[\frac{W_{t-1}^2 - 1}{\sqrt{2}}\right] = 0$$

Clearly,

$$\gamma_{t+h,t} = 0$$

for  $|h| \geq 2$ . Hence,

$$\gamma_{t+h,t} = \begin{cases} 1, & \text{if } h = 0, \\ 0, & \text{otherwise.} \end{cases}$$

Thus,  $\{X_t\} \sim wn(0, 1)$ . On the other hand, if  $t$  is odd,  $\{X_t\}$  and  $\{X_{t-1}\}$  are dependent so  $\{X_t\}$  is not

iid(0,1).

**b)** If  $n$  is odd

$$\mathbb{E}[X_{n+1}|X_1, X_2, \dots, X_n] = \mathbb{E}[W_{n+1}|W_0, W_2, W_4, \dots, W_{n-1}] = \mathbb{E}[W_{n+1}] = 0$$

and if  $n$  is even

$$\mathbb{E}[X_{n+1}|X_1, X_2, \dots, X_n] = \mathbb{E}\left[\frac{W_n^2 - 1}{\sqrt{2}}|W_0, W_2, W_4, \dots, W_n\right] = \frac{W_n^2 - 1}{\sqrt{2}} = \frac{X_n^2 - 1}{\sqrt{2}}.$$

This again shows that  $\{X_t\}$  is not iid(0,1).